Only The Best

SECO

CHUCKS

SINCE 1958

POWER CHUCKS • MANUAL CHUCKS • ROTARY CYLINDERS

An ISO 9001:2008 Company

POWER

OPEN CENTRE / HOLLOW DIRECT MOUNTING POWER CHUCK

TYPE OCD

3JAW

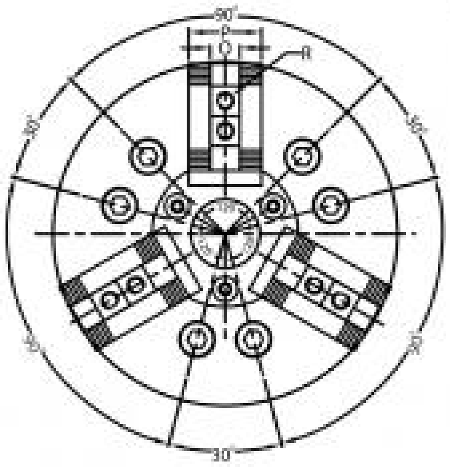
2JAW

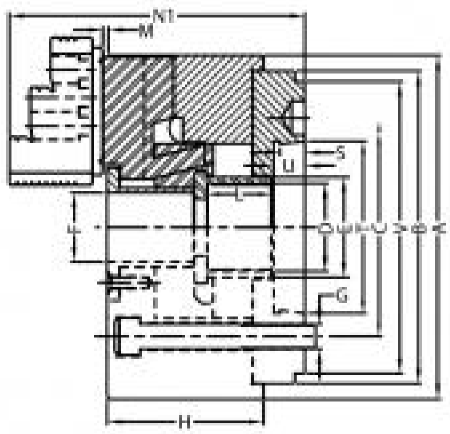
4JAW

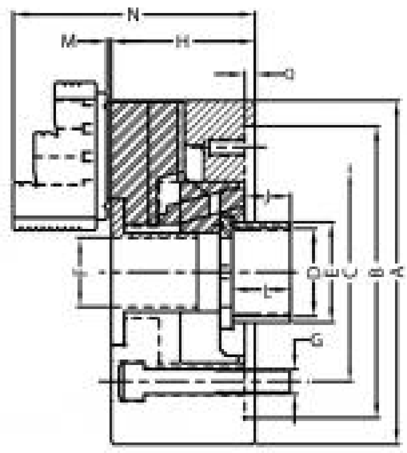
SECO open centre wedge hook chucks with thru hole are ideal for high speed chucking, bar chucking and universal machining. These chucks come fitted with back plate ready to mount on the machine.

SPECIAL FEATURES :

- High Speed
- Constant accuracy & high endurance
- Strong Gripping
- Available in 2, 3 & 4 Jaw configurations







A	B	Taper Nose	C	D	E	F	G	J Min.	J Max.	H	L	M	N	N1	O H7	P	Q	R	S	T	U	V	Serrations
135 mm	110	A2-4	82.6	M36 x 2.0	48	30	M10	-2.30	8.30	61	20	1.5	92	105.0	10	27	4	M8 x 1.25	10	65.513	12	108	1/16" x 90°
170mm	140	A2-5	104.8	M55 x 2.0	60	45	M10	-11.3	2.6	81	20	2	117	130.0	12	32	5	M10 x 1.25	12	82.563	15	133	1/16" x 90°
200 mm	170	A2-6	133.4	M55 x 2.0	62	48	M12	1.10	21.10	91	28	2	146	165.0	17	40	6	M12 x 1.75	13	106.375	16	165	1/16" x 90°
250 mm	220	A2-8	171.4	M72 x 2.0	79	65	M16	10.00	30.10	94	28	2	150	170.0	21	45	6	M16 x 2.0	14	139.719	18	210	1/16" x 90°
315 mm	220/ 300	A2-8/ A2-11	171.4/ 235	M92 x 2.0	102	80	M16/ M18	-20.00	22.00	91	28	4	153	177.0	21	50	6	M16 x 2.0	16	196.869	20	280	1/16" x 90°

LONG STROKE

A	B	Taper Nose	C	D	E	F	G	J Min.	J Max.	H	L	M	N	N1	O H7	P	Q	R	S	T	U	V	Serrations
135mm	110	A2-4	82.6	M36 x 2.0	48	33	3xM10	-2.3 0	8.2	71	20	1.5	102	115	10	27	4	M8 x 1.25	10	66.513	12	117	1.5mm x 60°
170mm	140	A2-5	104.8	M55 x 2.0	60	45	M10	-11.3	2.6	92	20	2	129	142	12	32	5	M10 x 1.25	12	82.563	15	133	1/16" x 90°
200 mm	170	A2-5	133.4	M55 x 2.0	62	48	M10	-8.20	22	100	28	2	154	173	17	40	6	M12 x 1.75	13	106.375	16	165	1/16" x 90°
210 mm	170	A2-5	133.4	M60 x 2.0	66	52	M12	-15.9	12.1	20	26	1	172	14	37	25	5	M12 x 1.75	13	106.375	16	145	1.5mm x 60°
250 mm	220	A2-8	171.4	M72 x 2.0	79	68	M16	-0	29	105	28	2	162	200	21	45	6	M16 x 2	14	139.719	16	210	1/16" x 90°
254 mm	220	A2-8	171.4	M85 x 2.0	78	75	M16	-8.5	23.5	112	20	2	168	172	16	42	6	M12 x 1.75	14	139.719	16	210	1.5mm x 60°
315 mm	220/ 300	A2-8/ A2-11	171.4/ 235	M92 x 2.0	98	80	M16	-0	32	103	28	4	165	200	21	50	6	M16 x 2	16	196.869	20	210/ 280	1/16" x 90°

SIZE	Max. RPM		Jaw Stroke (Per Jaw)		Plunger Stroke (Max.)		Weight (Kgs.) approx		Operating Force (Max. Kgf.)		Gripping Force (Max. Kgf.)	
	Normal	Long Stroke	Normal	Long Stroke	Normal	Long Stroke	Normal	Long Stroke	Normal	Long Stroke	Normal	Long Stroke
135 mm	6000	6000	2.75	4.5	10.0	20.0	10.0	11.0	1600	1600	7000	7000
170 mm	6000	6000	2.75	6.5	12.0	25.0	12.0	14.0	3000	3000	7000	7000
200 mm	5000	5000	3.75	7.5	17.0	30.0	21.0	24.0	4000	4000	9000	9000
210 mm	5000	5000	3.75	7.5	16.0	28.0	25.0	28.0	4000	4000	9000	9000
250 mm	4200	4200	4.4	7.5	19.0	30.0	35.0	38.0	5000	5000	10000	10000
254 mm	4200	4200	4.4	7.5	19.0	30.0	35.0	38.0	5000	5000	10000	10000
315 mm	3200	3200	4.4	8.0	19.0	30.0	50.0	55.0	6000	6000	13500	13500

NOTE : 1.5mm x 60° serrations are optional.

Chucks can be supplied with back plates of other International Standards on demand

SECO
POWER

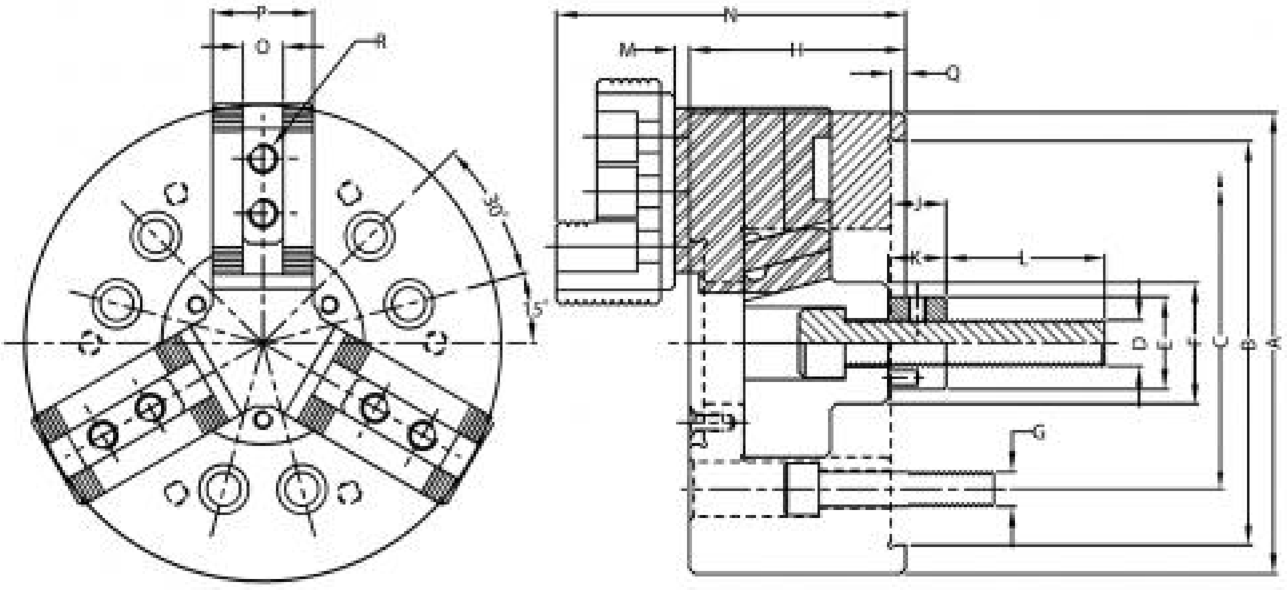
CLOSE CENTRE POWER CHUCK

TYPE CC

SECO close centre wedge hook type power chucks are cost effective solution for universal machining where thru-hole is not essential.

SPECIAL FEATURES :

- Constant accuracy & high endurance
- Strong Gripping
- Available in 2, 3 & 4 Jaw configurations



A	B H6	C	D	E	F	G	J		H	K	L	M	N	O H7	P	Q	R	SERRATIONS
							Min.	Max.										
125 mm	115.0	82.6	M12	24	35	M 10	22.0	39.0	59	20	55	9	100	10	25	4	M8 x 1.25	1/16" x 90°
160 mm	140.0	104.8	M16	32	42	M10	19.0	40.0	75	20	55	5	114	12	35	5	M10 x 1.5	1/16" x 90°
200 mm	170.0	133.4	M20	36	52	M12	19.0	41.0	90	20	55	6	148	17	40	6	M12 x 1.75	1/16" x 90°
250 mm	220	133.4 / 171.4	M20	36	52	M12 / M16	19.0	40.0	97	20	55	3	155	21	45	6	M16 x 2.0	1/16" x 90°
315 mm	220 / 300	171.4 / 235	M20	36	60	M16/ M18	16.0	42.0	105	20	50	4	171	21	45	6	M16 x 2.0	1/16" x 90°

NOTE : 1.5mm x 60° serrations are optional.

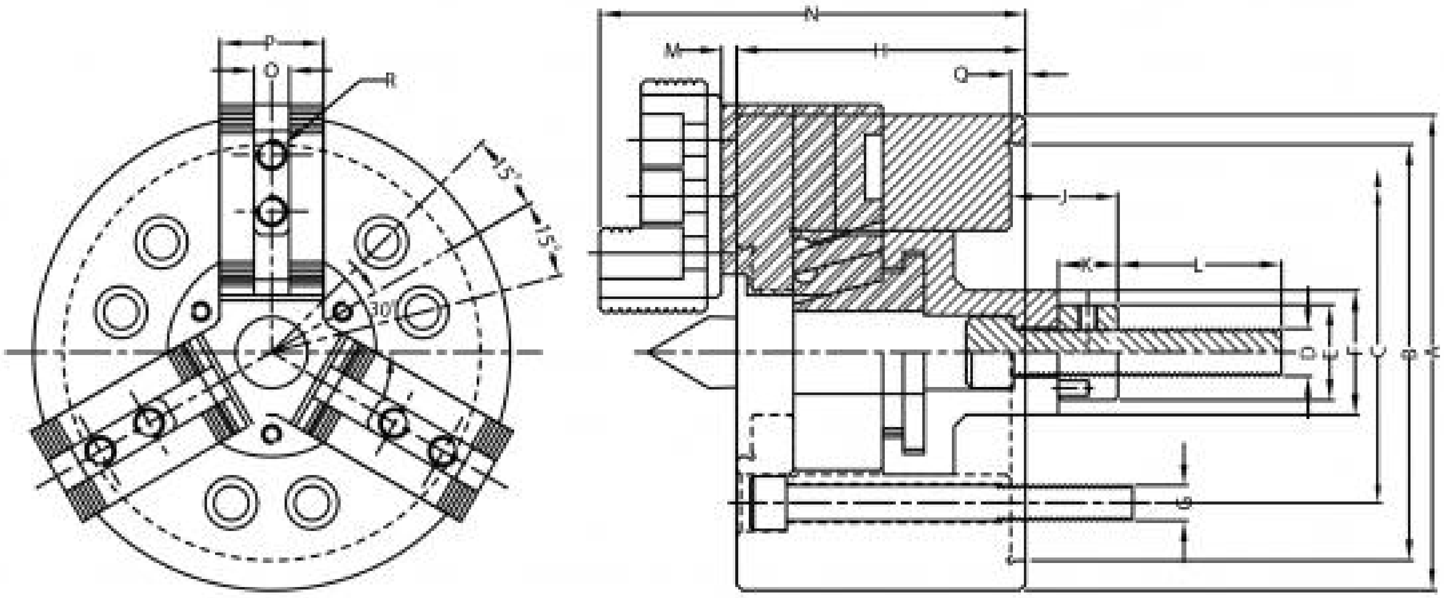
SIZE	Max. RPM	Jaw Stroke (Per Jaw)	Plunger Stroke (Max.)	Weight (Kgs.) approx	Operating Force (Max. Kgf.)	Gripping Force (Max. Kgf.)
125 mm	3000	3.5	17	8	600	1200
160 mm	4000	5	19	12	2000	4000
200 mm	3800	5	19	21	3500	7000
250 mm	2500	5	20	38	5000	10000
315 mm	2000	5	20	53	5000	10000

DIRECT MOUNTING OPTION AVAILABLE

SECO Jaw compensation chuck is ideal for between centre high speed, high load machining. The floating jaws while providing necessary grip does not dislocate the centre of the work piece. The chuck can easily be changed into an ordinary close centre chuck by replacing the centre retaining cartridge. Spring loaded centre also available.

SPECIAL FEATURES :

- Constant accuracy & high endurance
- Compensating Jaws
- Strong Gripping
- Available in 2, & 3 Jaw configurations



A	B H6	C	D	E	F	G	J		H	K	L	M	N	O H7	P	Q	R	SERRATIONS
							Min.	Max.										
160 mm	140.0	104.8	M16	32	42	M 10	37.0	55.0	97	20	55	5	140	12	35	5	M10 x 1.5	1/16" x 90°
200 mm	170.0	133.4	M20	36	52	M12	33.0	57.0	109	20	55	6	165	17	40	6	M12 x 1.75	1/16" x 90°
250 mm	220.0	133.4 / 171.4	M20	36	52	M12/M16	50.0	70.0	116	20	55	3	174	21	45	6	M16 x 2.0	1/16" x 90°
315 mm	220 / 300	171.4 / 235	M20	36	60	M16/ M18	48.0	68.0	120	20	50	4	178	21	45	6	M16 x 2.0	1/16" x 90°

NOTE : 1.5mm x 60° serrations are optional.

SIZE	Max. RPM	Jaw Stroke (Per Jaw)	Plunger Stroke (Max.)	Weight (Kgs.) approx	Operating Force (Max. Kgf.)	Gripping Force (Max. Kgf.)
160 mm	3000	5	19	17	2000	4000
200 mm	2800	5	19	26	3500	7000
250 mm	2200	5	19	45	5000	9000
315 mm	1800	5	19	60	5000	10000

DIRECT MOUNTING OPTION AVAILABLE

SOFT JAWS

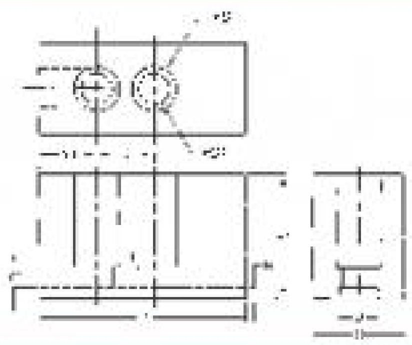
HARD JAWS

SOFT JAWS

(Steel & Aluminium)
Soft jaws for all kinds & makes of chucks both in 1.5 x 60° & 1/16 x 90° are available

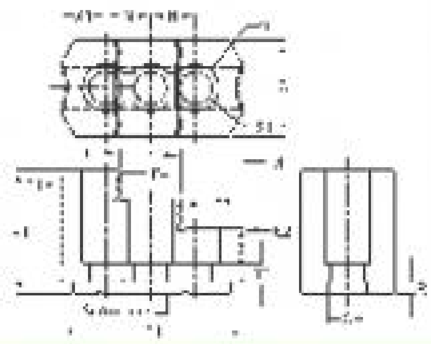
SOFT JAWS

SIZE	G	N	Q-H7	R	S	S1	T1	V1	X1	Y1	Serrations
125/135 mm	25	15	10	4.5	13.5	9.5	60	30	16	6	1/16" x 90°
165 mm	31	19	12	5	17.5	11.5	75	38	15	8	1/16" x 90°
170 mm	31	20	12	5	17.5	11.5	75	38	15	8	1.5" x 60°
200 mm	38	19	17	5	19.5	13.5	85	45	19	8	1/16" x 90°
210 mm	38	25	14	5	19.5	13.5	85	38	19	8	1.5" x 60°
250 mm	45	25	21	6	25.4	17.5	110	60	25	8	1/16" x 90°
254 mm	45	30	16	6	25.4	17.5	110	50	25	8	1.5" x 60°
305 mm	45	30	16	6	25.4	17.5	120	60	35	8	1.5" x 60°
315 mm	45	25	21	6	25.4	17.5	120	60	35	8	1/16" x 90°
380/450 mm	62	43	22	8	32.0	22.0	165	66	37	16	1/16" x 90°



TOP HARD JAWS

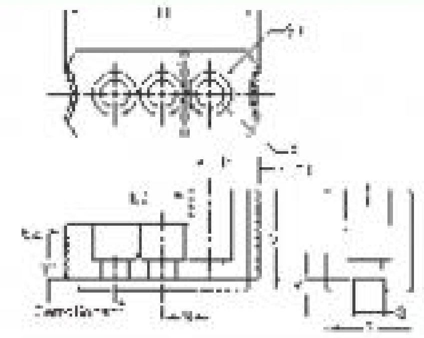
SIZE	G	N	Q-H7	R	S	S1	T1	V1	X1	Y1	Serrations
125/135 mm	25.5	15	10	4.5	13.5	9.5	61	30	11.5	6	1/16" x 90°
165 mm	31	19	12	5	17.5	11.5	68.5	35	13.5	8	1/16" x 90°
170 mm	31	20	12	5	17.5	11.5	70	35	13.5	4	1.5" x 60°
200 mm	40	19	17	5	19.5	13.5	77	52	19	8	1/16" x 90°
210 mm	40	25	14	5	19.5	13.5	85	51	16.5	8	1.5" x 60°
250 mm	45	25	21	6	25.4	17.5	95	55	24	8	1/16" x 90°
254 mm	40	30	16	5	19.5	13.5	100	54	20	8	1.5" x 60°
305 mm	45	30	16	5	25.4	17.5	103	59	21.5	10	1.5" x 60°
315 mm	45	25	21	6	25.4	17.5	103	58	32	8	1/16" x 90°
380/450 mm	62	43	22	8	32.0	22.0	149	86	32	13	1/16" x 90°



CLAMPING RANGE

SIZE	E1	E2	E3	E4	I1	I2	I3
125/135 mm	15-60	15-30	80-125	-	60-115	-	-
165 mm	22-72	22-40	80-129	-	62-136	-	-
200 mm	25-99	25-102	74-152	124-200	70-150	116-190	166-200
250 mm	28-104	41-124	123-206	205-250	76-160	156-242	236-250
315 mm	42-170	56-195	135-275	220-315	84-218	236-250	228-315
380 mm	50-200	50-195	175-320	245-385	155-295	220-365	50-195
450 mm	50-260	50-260	175-385	245-450	155-360	220-440	50-260

NOTE : Any jaws can be supplied with serrations 1.5" x 60° or 1/16" x 90° as a special case



The advancement of technology has enabled us to produce machines with ultra high speed. The high speed cutting technology has posed new challenges for the chucking equipment. The work holding at higher speeds needs chucking equipment with the competent gripping force.

The gripping force of the chuck depends upon various factors such as:

- Lubrication conditions
- Size, shape and finish of the job
- Gripping surface of the jaws
- Draw bar pull
- Centrifugal forces applied at higher speed
- The Weight of top and base jaws

Each of the above factors has its own role in determining the gripping force of the Chuck. However the influence of centrifugal force on gripping force of a chuck needs better understanding by the user so that he can select cutting speeds and feeds precisely for optimum productivity.

It is advisable for the chuck user to understand the following formula for calculating centrifugal force:

$$C.F. = \frac{0.102 \times G \times r \times (\pi \times n)^2}{30}$$

Where ,

G = weight of base jaws and top jaws
r = the radius of center of gravity
n = Operating speed
C.F. = centrifugal force

After having calculated the centrifugal force, the available chucking force at a given speed can be found with the help of following equation : -

$$GF_1 = GF_2 \pm CF$$

Where ,

GF1 = Gripping force at a given speed
GF2 = Gripping force when the chuck is in a stationary stage
CF = Centrifugal force.

+(Plus) for internal gripping and – (minus) for external gripping.

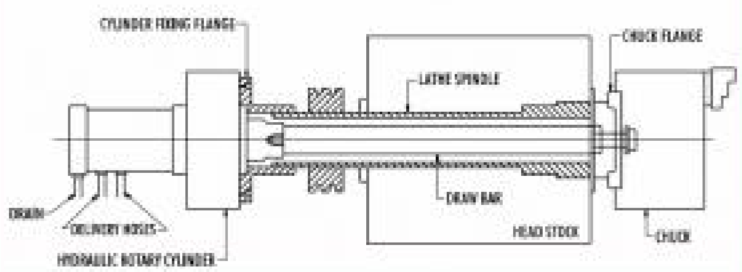
It is advisable that the chucking force at a given speed should be at least 1.5/2.0 times the cutting forces depending upon the constructional rigidity/fragility of the job.
For more details see operating manual

Elements of Power Chucking System

Power chucking system consists of

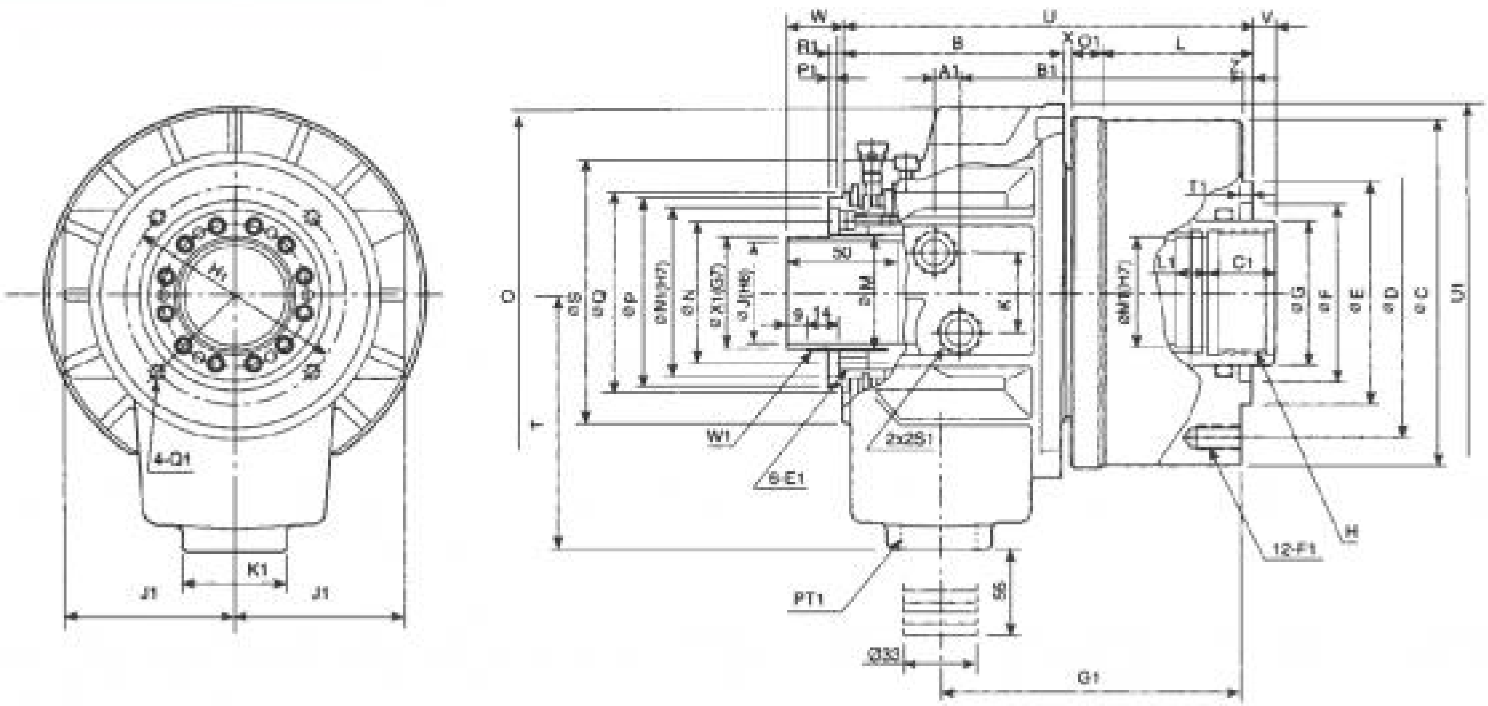
- Power Chuck.
- Power source (Hydraulic or Pneumatic Revolving Cylinder).
- Draw Bar.
- Power Pack.
  Schematic arrangement of a hydraulic Chuck and Cylinder is shown in the figure.

MOUNTING OF CYLINDER & CHUCK



SPECIAL FEATURES

1. LIGHT WEIGHT
2. LARGE BORE
3. HIGH SPEED
4. SMALL & COMPACT
5. DYNAMICALLY BALANCED
6. IN BUILT SAFETY MECHANISIM
   WITH SAFETY CHECK VALVES



DIMENSIONS

Model	C1	E1	F1	G1	H1	J1	K1	L1	M1	N1	O1	P1	Q1	R1	S1	T1	U1	W1	X1	B	C
OBC 100	25	M5x8	M10x1.5	126	88	68	53	15	38	64	14	4	M65x0.8	46	PT3/8"	6	136	M44x1.5	42	101	136
OBC 120	30	M6x1.0	M10x1.5	135	98	76	47	15	50	76	14	4	M5x0.8	6	PT1/2"	6	169	M52x1.5	50	99	154.5
OBC 160	30	M6x1.0	M10x1.5	145	110	86	47	15	55	85	14	4	M6x1.0	7	PT1/2"	6	187.5	M58x1.5	56	105	190
OBC 180	35	M6x1.0	M10x1.5	116.5	155	101	47	15	80	108	16	4	M6x1.0	7	PT1/2"	6	220	M84x2	81	126	215
OBC 200	350	M6x1.0	M12x1.75	182	165	110	47	15	95	120	16	4	M6x1.0	7	PT1/2"	6	267	M99x2	96	141	230

Model	D	E	F	G	H	J	K	L	M	N	O	P	Q	S	T	U	V max	V mix	W max	W mix	x	z	A1	B1
OBC 100	118	100	65	50	M42x2	38	32	62	49.8	54	196	73	80	104	115	179.5	10	-5	39	24	2.5	5	11	120.5
OBC 120	130	100	80	65	M55x2	46	36	67	52.6	64	166	85	90	118	114	184	10	-5	40	25	4	5	11	126.5
OBC 160	170	130	85	70	M60x2	52	36	75	59.6	73	184	96	102	137	130	196	17	-5	47	25	4	5	11	136
OBC 180	190	160	120	95	M85x2	75	36	84	84.6	98	215	121	131	166	160	230	20	-5	50	25	4	5	17.5	153.5
OBC 200	190	160	125	110	M98x2	91	36	84	97.6	108	265	138	148	182	185	253	25	-5	55	25	4	5	15	116.5

SECD POWER

OPEN CENTRE (Through Hole)
SUPER HIGH SPEED ROTARY HYDRAULIC CYLINDER

TYPE OBC

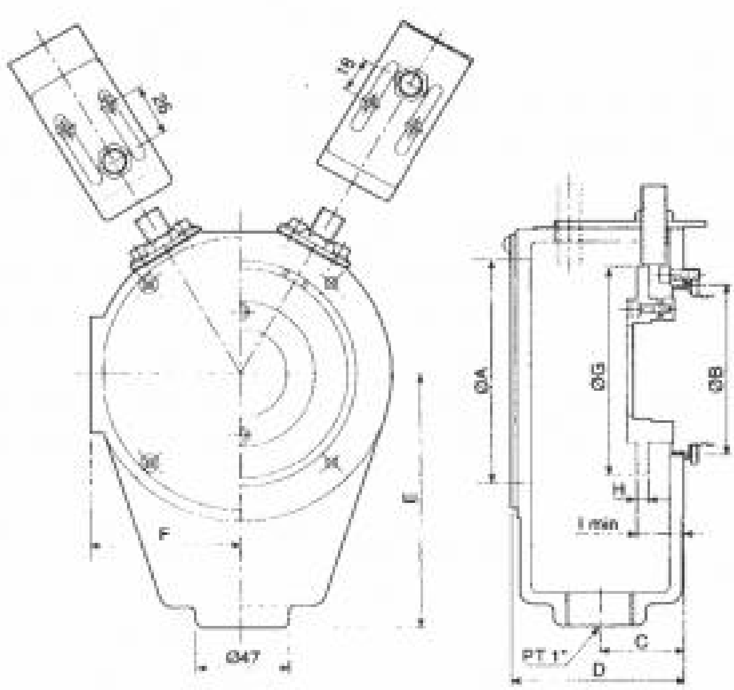
SPECIFICATIONS

Model	Piston Dia (mm)	Push Side (cm^2)	Pull Side (cm^2)	Piston Stroke (mm)	Push Side KN(kgf)	Pull Side KN(kgf)	Max. Operating Pressure (kgf/cm^2)	Max. Speed (r.p.m)	Moment Inertia 1($kgf.m^2$)	Weight in (Kg)	Total Leakage (*L*/min)
OBC 100	100	71	68.5	25	24(2529)	24(2447)	40.8	8000	0.011	8.6	3.0
OBC 120	125	100	89	25	38(3875)	33(3365)	40.8	7000	0.019	12.0	3.0
OBC 160	155	161	150	22	60(6118)	56(5710)	40.8	6200	0.052	16.8	3.9
OBC 180	180	198	183	25	75(7546)	69(7036)	40.8	4700	0.095	26.0	4.2
OBC 200	200	252	234	25	94(9585)	88(8973)	40.8	3800	0.015	37.0	4.5

SECD POWER

OPEN CENTRE HOLLOW (Through Hole)
HYDRAULIC CYLINDER COOLENT COLLACTORS

TYPE CLC

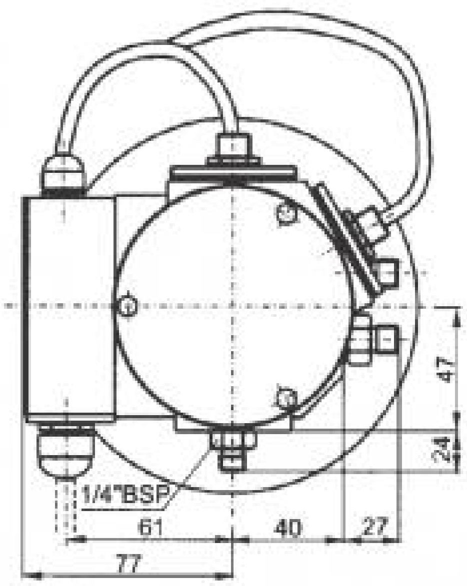


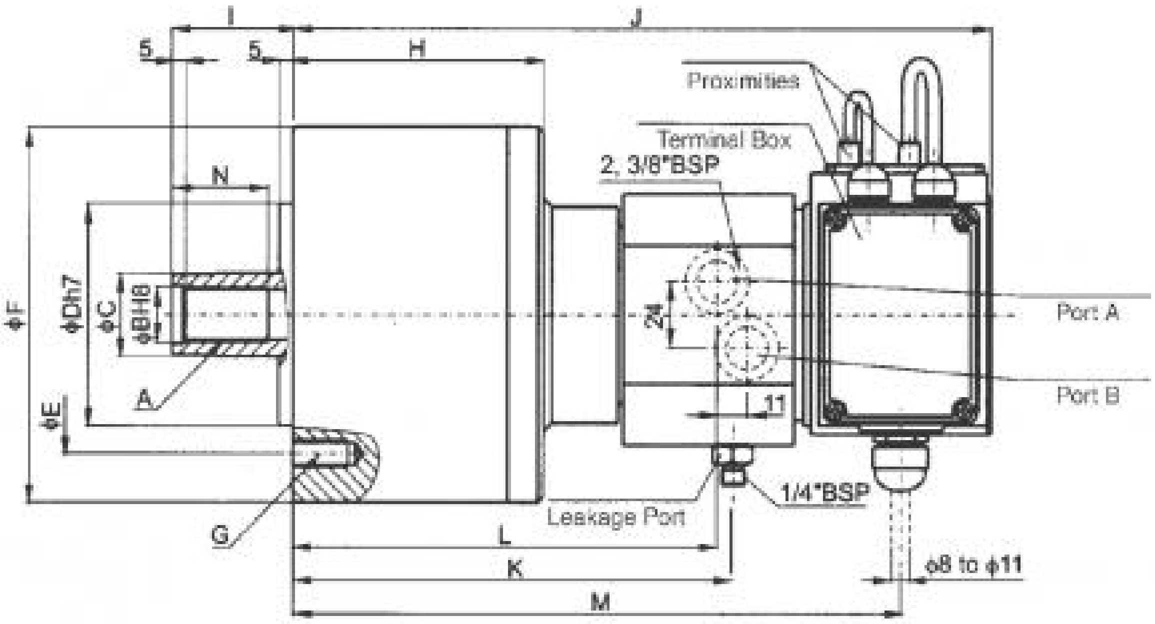
DIMENSIONS

Model	A	B	C	D	E	F	G	H	I min	Matching Cylinder
CLC 120	118	87	42	84	131	76	107	6	23	OBC120
CLC 160	158	98	44	88.5	151	96	147	6	23	OBC160
CLC 180	158	123	44	88.5	151	96	147	6	23	OBC180

SPECIAL FEATURES

1. COMPACT DESIGN
2. HIGH SPEED
3. DYNAMICALLY BALANCED
4. IN BUILT SAFETY MECHANISIM
   WITH SAFETY CHECK VALVES





DIMENSIONS

Model	A mm	ø B mm	øC mm	øDh7 mm	øE mm	F mm	G mm	H mm	I max mm	I min mm	J mm	K mm	L mm	M mm	N mm	O mm	P mm	øQ mm
HCC 125	M24x3.0	25H8	35	110	130	162	6-M12x24	101	51	26	261.5	166	162	227	44	6-M10x105	15	142
HCC 150	M30x3.5	31H8	45	110	130	190	12-M12x24	106	56	26	266.5	171	167	232	45	6-M10x100	15	170

SPECIFICATIONS

Model	Max. Pull kg.	Pull Area (cm^2)	Piston Dia mm	Piston Travel (mm)	Max. Speed (r.p.m)	Max. Leakage at 30 Bar and 50ºc lit/min	Weight in (Kg)	GD^2 $kg\text{-}m^2$
HCC 125	4000	113	125	25	6000	0.8	12	0.0264
HCC 150	5700	160	150	30	5500	0.8	15	0.0530

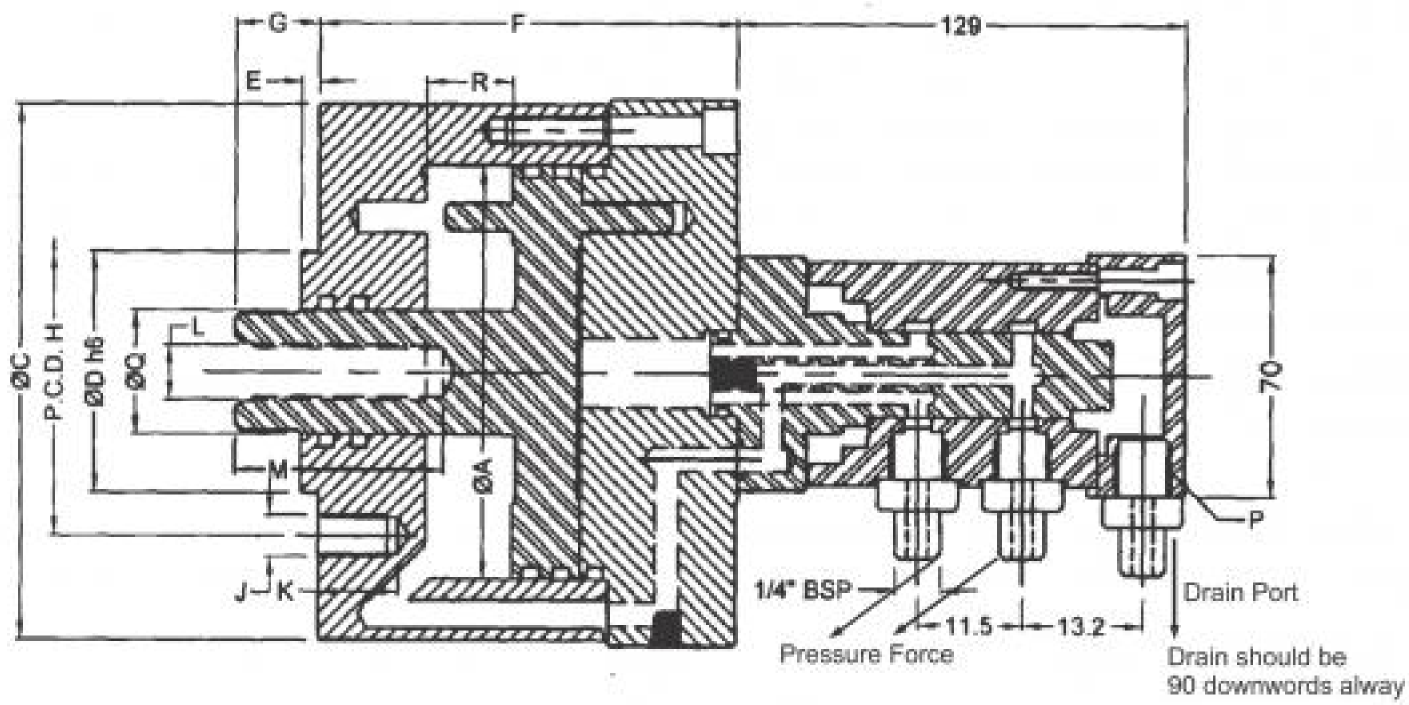
Proximity Switches Optional

POWER

Hydraulic Close Centre ROTATING CYLINDERS

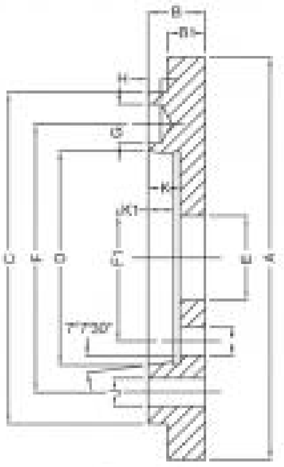
TYPE CBC

SECO Close Centre Rotating Cylinders
Suitable for CNC and AUTO chucking lathes.
Years of trouble free service assured



DIMENSIONS

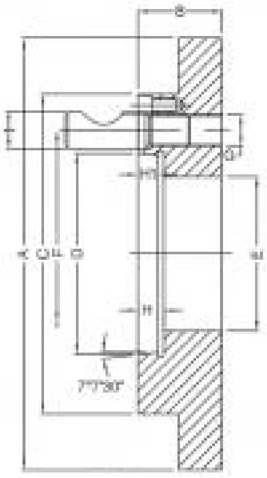
Model	AØ	CØ	D-h6	E	F	G Min.	G Max.	H P.C.D	J	K	L	M	P	Q	R	Wt. in kgs. App.	Draw Bar Pull in Kgs. at 25 Kg/cm^2	RPM
CBC 60	60	95	60	5	84	24	49	75	4 x M8	20	M12	50	3/8"BSP	25	25	5	1000	4000
CBC 90	90	125	70	5	121	24	49	95	4 x M10	20	M14	50	3/8"BSP	30	25	6.8	1400	4000
CBC 120	120	160	70	5	121	24	49	95	4 x M12	20	M16	50	3/8"BSP	36	25	9	2600	4000
CBC 160	160	200	95	5	126	24	54	145	4 x M16	25	M24	50	3/8"BSP	36	30	13.8	4800	3000
CBC 200	200	240	125	5	126	24	54	170	6x M16	25	M24	50	3/8"BSP	36	30	20	7600	2500



A-1, A-2

A1, A2 TYPE

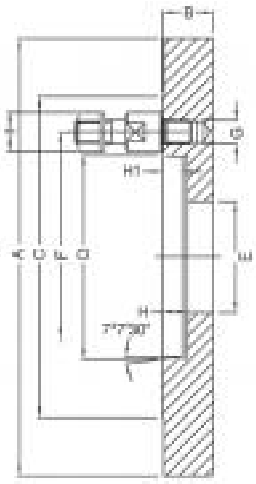
SIZE	4	5	6	8	11	15
A	160, 200, 250	160, 200, 250	200, 250, 315	250, 315, 400	315, 400, 600	315, 400, 500, 630
B	33.0	35.00	35.0	45.0	45.0	50.0
C	117	146	165	210	280	380
D	63.513	82.563	106.375	139.719	196.869	285.775
E (Max)	61.0	79.0	103.0	136.0	193.0	282.0
F (A2)	82.6	104.8	133.4	171.4	235.0	330.2
F1 (A1)	-	61.90	82.6	111.10	165.10	247.6
G	14.7	16.3	19.45	24.20	29.4	35.7
H	6.5	6.5	6.5	8.0	10.0	10.0
J	12.0	12.0	14.0	18.0	20.0	24.0
K	-	15.0	16.0	18.0	20.0	21.0
K1	10.0	12.0	13.0	14.0	16.0	17.0



D-1

D1 TYPE

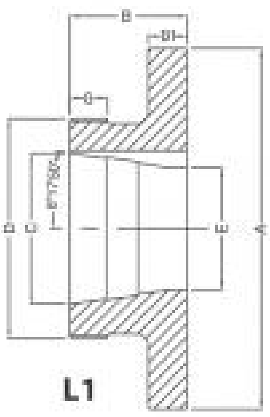
SIZE	4	5	6	8	11	15
A	160, 200, 250	160, 200, 250	200, 250, 315	250, 315, 400	315, 400, 600	400, 500, 630
B	33.0	35.00	40.0	45.0	50.0	55
C	117	146	181	222	298	403
D	63.513	82.563	106.375	139.719	196.869	285.775
E (Max)	61.0	79.0	103.0	136.0	193.0	282.0
F	82.6	104.8	133.4	171.4	235.0	330.2
G	M-10x1	M-12x1	M-16x1.5	M-20x1.5	M-22x1.50	M-24x1.5
H	13	15	16	18	20	21
H1	10	12	13	14	16	17
I	15.9	19	22.2	25.4	30.2	34.9



DIN

DIN TYPE

SIZE	4	5	6	8	11	15
A	160, 200, 250	160, 200, 250	200, 250, 315	250, 315, 400	315, 400, 600	400, 500, 630
B	33.0	35.00	40.0	45.0	50.0	55
C	117	146	165	222	290	380
D	63.525	82.575	106.39	139.735	196.885	285.80
E (Max)	61.0	79.0	103.0	136.0	193.0	282.0
F	85	104.8	133.4	171.4	235.0	330.2
G	M-10x1	M-10x1	M-12x1	M-16x1	M-20x1	M-24
H	13	16	17	19	21	22.0
H1	10	12	13	14	16	18.0
I	19.5	19.5	21.5	27	34	-



L1

L1 TYPE

SIZE	A	B	B1	C	D	E	G
L0	160, 200, 250, 315	63.5	28	82.55	4.1/4x6	66	15
L1	200, 250, 315, 400	76.2	28	104.78	6x6	84.5	16

A1,A2

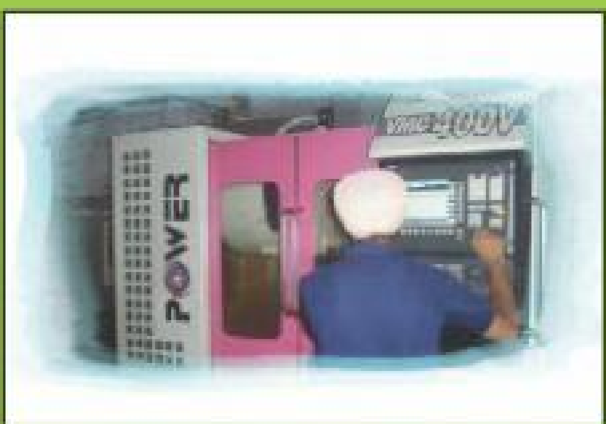
D1

DIN
55027/22

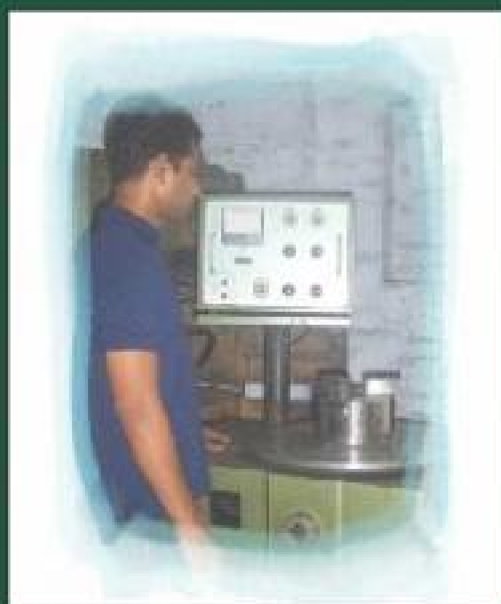
L1

MODERN TECHNOLOGY

IN HOUSE CNC MACHINING



DYNAMIC GRIP TEST



DYNAMIC
BALANCING MACHINE

Face Plate Jaws suitable for heavy duty gripping jobs on vertical turning lathes, boaring mills & face plates

MATERIAL COMPOSITION:

BODY

C I BODY : High tensile, wear resistant cast iron, Hardness180-210 Bhn.

JAWS

Case hardened Precision ground
Serrated faces provide extra grip.

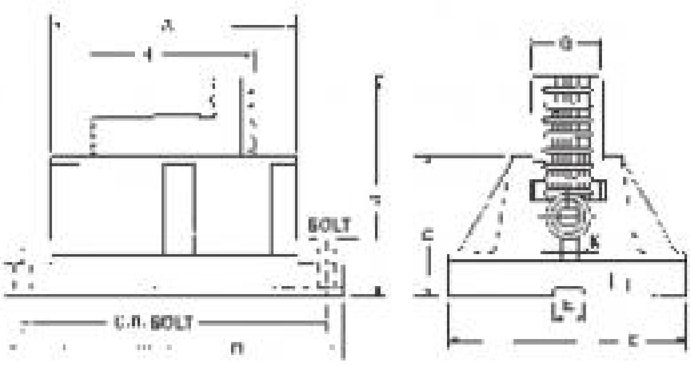
Screw & Lock

Case Hardened alloy steel. Interchangeable Parts.

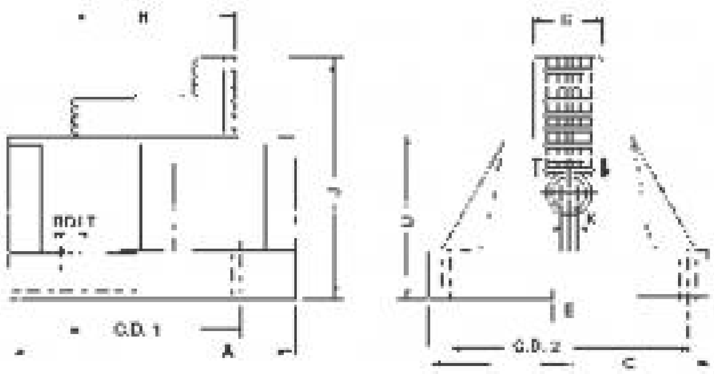
DIMENSIONAL DATA

Single Type	A	B	C	D	E	Bolt	C.D. Bolt	G	H	J	K	Wt. in Kg.
S - 150	155	240	140	115	23	M-12	200	55	132	174	14	17
S - 200	205	290	140	115	23	M-12	251	55	140	174	14	20

Single Type

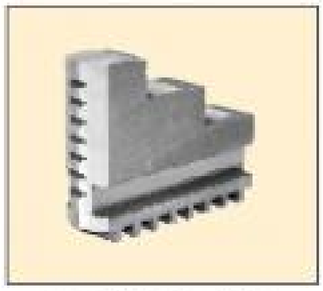


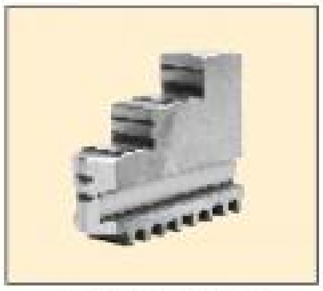
Double Type



Double Type	A	C	D	E	Bolt	C.D. 1	C.D. 2	G	H	J	K	Wt. in Kg.
D - 150	150	200	115	23	M-12	120	170	55	135	180	14	18
D - 200	200	200	115	23	M-12	120	170	55	135	180	14	23

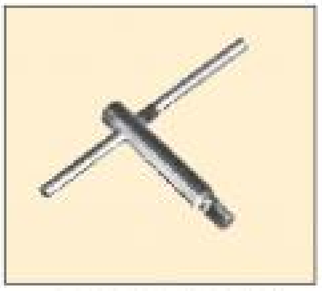
FOR
SJ, MTJ, AS & DP CHUCKS

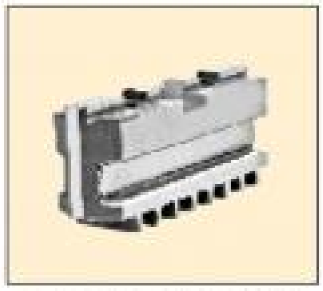

FORWARD JAW


REVERSE JAW

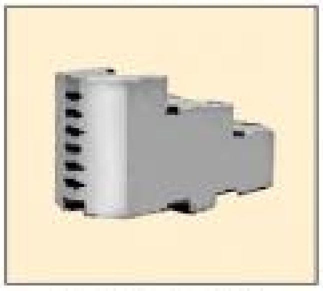

PINION & PIN


SCROLL


OPERATING KEY


BASE/MASTER JAW


TOP SOFT JAW


TOP HARD JAW


BASE JAW DP CHUCK


TOP JAW DP CHUCK


ADJUSTING SCREW
For A S CHUCK


HEMISPHERE BALL
For AS CHUCK

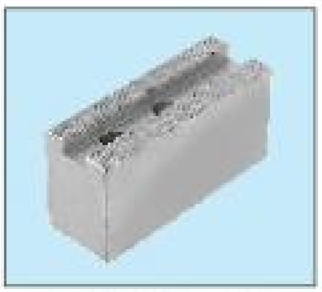

BACK COVER PLATE

FOR
POWER CHUCKS


BASE JAW


WEDGE


TOP HARD JAW


TOP SOFT JAW


SPINDLE NUT


SPINDLE NUT PLATE

FOR
BACK PLATES


CAM LOCK

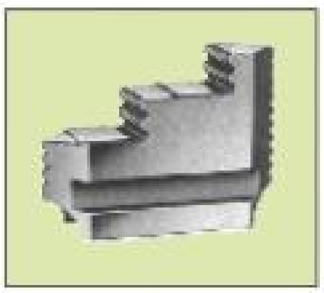

BACK PLATE STUD
For DIN 55027/22


CAM LOCK STUD

FOR
INDEPENDENT CHUCKS


THRUST BEARING


OPERATING SCREW


HARD REVERSIBLE JAWS


BASE HARD JAW

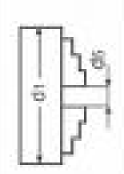
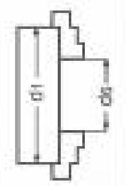
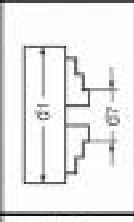
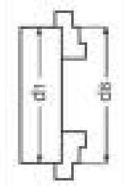
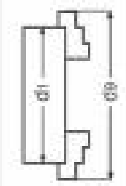
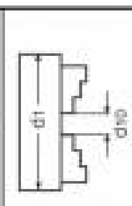
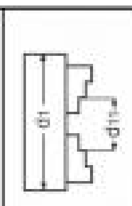
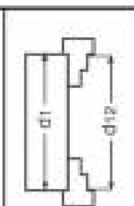
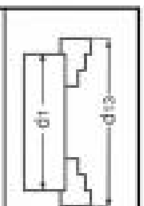
MAXIMUM SPEED For Manual Self Centring Chucks

CHUCK SIZE	CAST IRON BODY	S.G IRON BODY	STEEL BODY
80 mm	4000	5600	7000
100 mm	3300	4800	5500
125 mm	2700	3800	4500
160 mm	2000	2500	3200
200 mm	1600	2200	2800
250 mm	1300	2000	2400
315 mm	1000	1500	1900
400 mm	850	1200	1500
500 mm	650	1000	1200
630 mm	545	800	1000

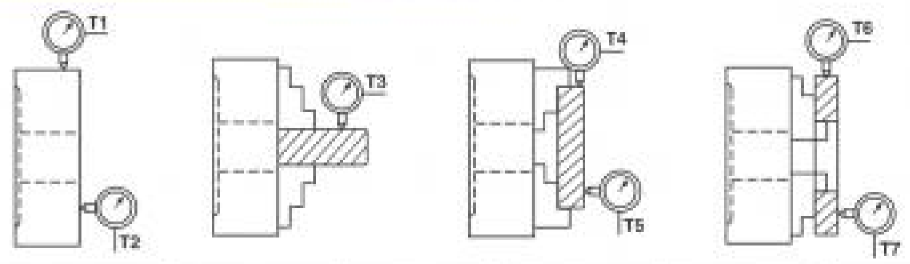
CAUTIONS :-

- It is better to use chucks at75% of the maximum RPM.
- Gripping force can drop due to centrifugal force if jaws are changed by user. In such cases users should be access the speed or consult us.
- Older chucks should be operated at low RPM as there is possibility of loss of gripping force due to wear & tear.

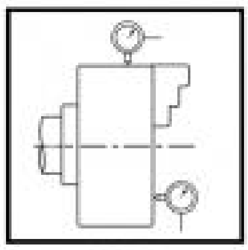
CLAMPING RANGE of 3 Jaws Manual Self Centring Chucks

SIZE									
d1	d5min.	d6max.	d7min.	d8max.	d9max.	d10min.	d11min.	d12max.	d13max.
80	2	27	22	69	91	2	25	71	91
100	3	34	28	87	115	3	31	93	115
125	3	46	35	112	148	3	33	120	148
160	3	62	43	154	200	3	48	160	200
200	4	84	50	194	250	4	56	200	250
250	5	108	66	240	310	5	70	250	310
315	10	145	70	299	385	10	90	315	385
400	20	198	85	380	480	20	125	400	480
500	35	280	110	480	600	35	132	500	600
630	50	390	130	610	750	50	190	630	750

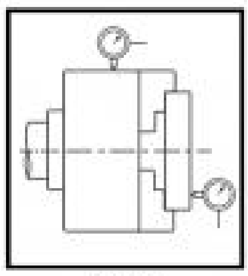
TEST SPECIFICATIONS of Manual Self Centring Chucks



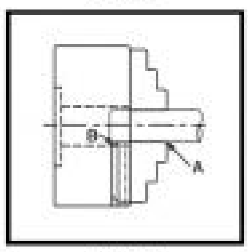
Chuck Dia in MM	T1,T2	T3	T4	T5	T6	T7
80-125	0.025	0.075	0.04	0.025	0.04	0.025
160-250	0.03	0.04	0.04	0.03	0.04	0.03
315-400	0.04	0.075	0.06	0.04	0.06	0.04
500-630	0.06	0.10	0.075	0.05	0.075	0.05



TEST-A

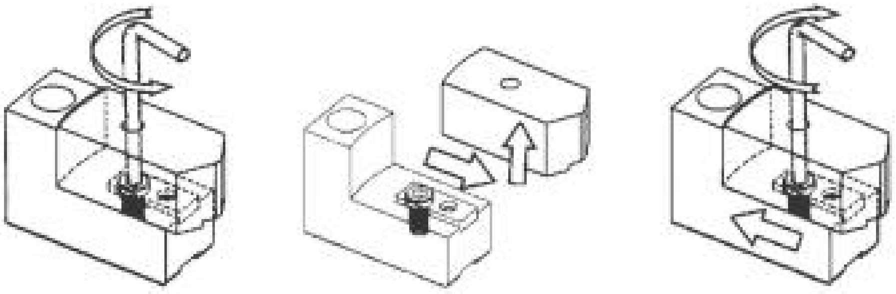


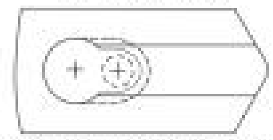
TEST-B



TEST-C

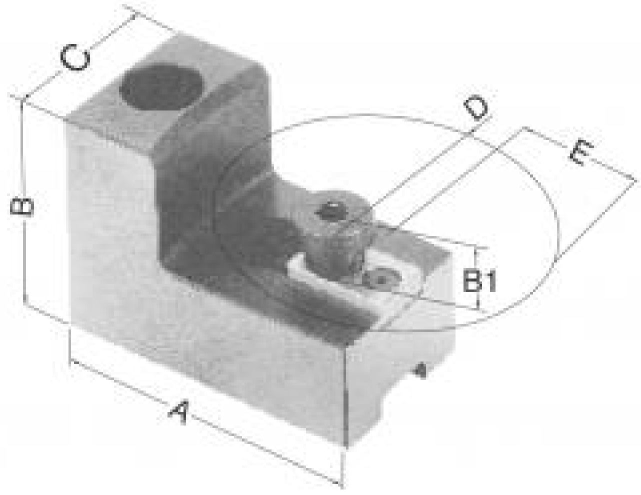
NOMINAL SIZE CHUCK		TEST -A	TEST-B	TEST-C
OVER	UPTO & INCLUDING	PERMISSIBLE DEVIATION	PERMISSIBLE DEVIATION	PERMISSIBLE DEVIATION
----	200	0.050	0.050	.04 FILLER GAUGE SHALL NOT PASS AT POINT A & POINT B
200	400	0.100	0.050	
400	630	0.125	0.125	
630	800	0.150	0.150	
800	1000	0.200	0.200	





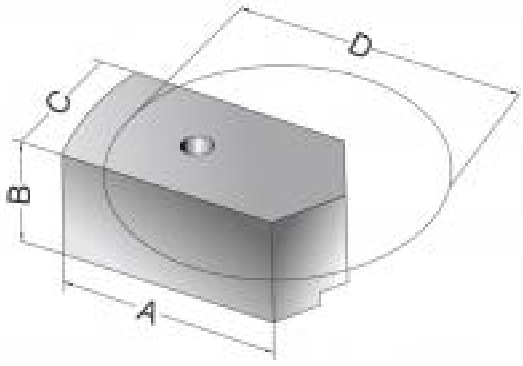
BASE JAWS

CHUCK SIZE	Length A	Height B	Height to Screw B1	Width C	Max Insert Rad. D	Base Jaw E
165 mm	60	44	17.45	31.75	63.5	47
200 mm	74	54.5	17.45	38.1	82.55	58
250 mm	79	54.5	17.45	44.45	98.5	89.7
315 mm	90	65.65	17.45	50.8	120.65	117.3
380 mm	127	67.20	17.45	63.5	158.75	61



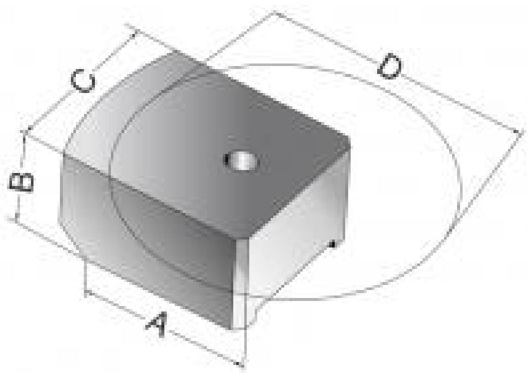
SOFT KEY INSERT JAWS

CHUCK SIZE	STANDARD HEIGHT				
	Length A	Height B	Width C	Bore Dia D	Extra Height B
165 mm	54.25	25.4	31.75	101.6	50.8
200 mm	73.53	31.75	38.1	139.7	57.15
250 mm	89	31.75	44.45	176.53	57.15
315 mm	110.9	38.1	50.8	215.9	63.5
380 mm	127	38.1	63.5	292.1	63.5



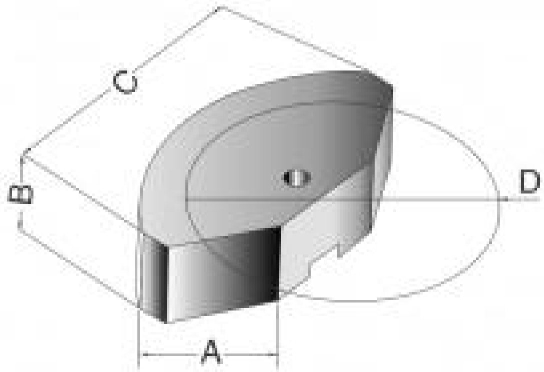
STRADDLE INSERT JAWS

CHUCK SIZE	STANDARD HEIGHT			
	Length A	Height B	Width C	Bore Dia D
165 mm	45.72	25.4	44.45	101.6
200 mm	62.5	31.75	50.8	139.7
250 mm	76.2	31.75	57.15	171.45
315 mm	96.2	38.1	63.5	215.9
380 mm	127	44.45	76.2	292.1



STRADDLE FULL GRIP INSERT JAWS

CHUCK SIZE	STANDARD HEIGHT			
	Length A	Height B	Width C	Bore Dia D
165 mm	45.72	25.4	76.2	101.6
200 mm	62.5	31.75	101.6	139.7
250 mm	76.2	31.75	127	171.45
315 mm	96.2	38.1	152.4	215.9
380 mm	127	44.45	190.5	292.1



SECO

CHUCKS

Manufacturers

STATE ENGINEERING CORPORATION

SANT NAGAR, CHACHOKI, PHAGWARA - 144 632 (Pb.) INDIA

Tel. : +91-1824-261677, 223392 Fax. : +91-18 24-260887

Email : secochucks@gmail.com Web. : http://www.secochuck.com

CE ISO 9001:2008 DNV

art masters
094641-85007

CAT 2015/1

SECO GOLD

CONTENTS

STANDARD JAWS SELF CENTRING CHUCKS

TYPE SJ

SJ3

SJ4

SJ6

Scroll operated self centring chuck with 2 sets of jaws (Reverse and Forward) is widely used in mass production.

MATERIAL COMPOSITION:

BODY

C I BODY : High tensile, wear resistant cast iron alloyed with Chrome and Manganese, Hardness180-210 Bhn.

SG IRON BODY : 500/7 grade of SG Iron, Hardness 180-210 Bhn

STEEL BODY : Medium carbon steel of 600 mpa tensile strength

JAWS

Case hardening steel with tough core, hardness 56-60 Hrc.

PINION

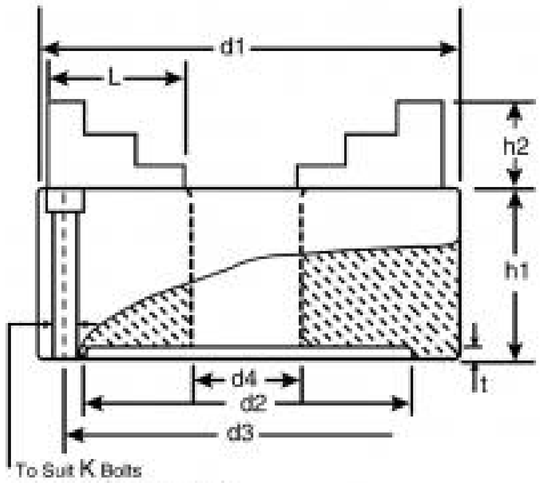
Case hardening alloy steel hardness 42-50 Hrc.

SCROLL

Forged high strength wear resistant case hardening alloy steel, Hardness 40-45 Hrc.

SPECIAL FEATURES

- BODY : Jaw ways Hardened and ground.
- JAWS : Jaws teeth Precision ground from both sides
- SCROLL : Scroll threads precision ground from both sides. Hardness 40-45 Hrc.
- PARTS : Interchangeable
- RUNOUT : Less than 0.04 from 160 mm onwards



To Suit K Bolts
3 & 6 Jaw Chucks 3 Holes
4 Jaw Chucks 4 Holes

DIMENSIONS IN MM

Nominal Size d_1	d_2 H_7	P.C.D d_3	P.C.D Tol. on d_3	d_4 Min.	h_1 Max.	h_2 Max.	t Min.	L	K	Size of Sq. Key for Chuck	Large Bore	Approx. Wt. in Kgs. SJ3	SJ4	SJ6
80	56	67	± 0.2	16	44	12	3	32	M6	6	-	1.6	2.0	-
100	70	83	± 0.25	20	50	18	3	41	M8	8	-	2.5	3.0	-
125	95	108	± 0.25	33	57	21	4	48	M8	9	-	4.5	5.0	6.0
160	125	140	± 0.3	42	67	29	4	65	M10	10	-	8.0	9.0	10.0
200	160	176	± 0.3	55	75	31	4	80	M10	11	-	14.0	15.0	16.0
250	200	224	± 0.35	76	85	36	5	98	M12	12	-	27.0	28.0	30.5
315	260	286	± 0.4	107	87	36	5	118	M12	14	-	38.5	41.0	44.0
400	330	362	± 0.4	150	110	51	5	160	M16	16	175.0	87.0	88.0	93.0
500	420	458	± 0.4	200	120	65	7	194	M16	16	225.0	150.0	162.0	223.0
630	545	586	± 0.4	275	135	65	7	248	M16	19	335.0	246.0	307.0	338.0

CHUCKS WITH BACK PLATE ALSO AVAILABLE

GOLD

MASTER GRIPPING SELF CENTRING CHUCKS

TYPE MTJ

MTJ3

MTJ2

MTJ4

MTJ6

The two piece design of jaws of this chuck offers wide range of application.Different sets of top jaws, suitable for different jobs mountable on some base jaws are solution to numerous odd problems. Standard provision of top hard reversible set and soft set of jaws make this chuck versatile for precision machining and mass production.

SPECIAL FEATURES

BODY : Knob & Jaw ways Hardened and ground.

JAWS : Jaws teeth Precision ground from both sides

SCROLL : Scroll threads precision ground from both sides. Hardness 40-45 Hrc.

PARTS : Interchangeable

RUNOUT : Less than 0.04 from 160 mm onwards

MATERIAL COMPOSITIONS :

BODY

C I BODY : High tensile, wear resistant cast iron alloyed with Chrome and Manganese, Hardness180-210 Bhn.

SG IRON BODY : 500/7 grade of SG Iron, Hardness 180-210 Bhn

STEEL BODY : Medium carbon steel of 600 mpa tensile strength

JAWS

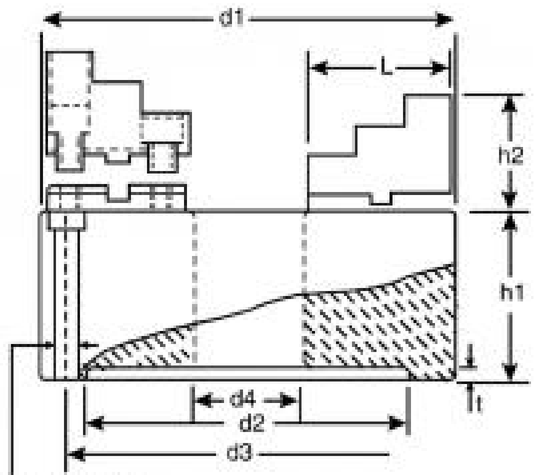
Case hardening steel with tough core, hardness 56-60 Hrc.

PINION

Case hardening alloy steel hardness 42-50 Hrc.

SCROLL

Forged high strength wear resistant case hardening alloy steel, Hardness 40-45 Hrc.



To Suit K Bolts
3 & 6 Jaw Chucks 3 Holes
2 & 4 Jaw Chucks 4 Holes

DIMENSIONS IN MM

Nominal Size d_1	d_2 H_7	P.C.D d_3	P.C.D Tol. on d_3	d_4 Min.	h_1 Max.	h_2 Max.	t Min.	L	K	Size of Sq. Key for Chuck	Large Bore	Approx. Wt. in Kgs. 3JAW	4JAW	6JAW
125	95	108	$\pm$0.25	33	57	37	4	53	M8	9	-	5.0	5.5	-
160	125	140	$\pm$ 0.3	42	67	44	4	72	M10	10	-	8.0	9.0	10.0
200	160	176	$\pm$ 0.3	55	75	52	4	84	M10	11	-	15.0	18.0	27.0
250	200	224	$\pm$0.35	76	85	61	5	103	M12	12	-	29.0	30.0	32.0
315	260	286	$\pm$ 0.4	107	87	63	5	123	M12	14	-	42.0	48.0	50.0
400	330	362	$\pm$ 0.4	150	110	77	5	165	M16	16	175.0	90.0	91.0	96.0
500	420	458	$\pm$ 0.4	200	120	65	7	194	M16	16	225.0	165.0	182.0	253.0
630	545	586	$\pm$ 0.4	275	135	65	7	248	M16	19	335.0	264.0	331.0	374.0

CHUCKS FITTED WITH BACK PLATE ALSO AVAILABLE

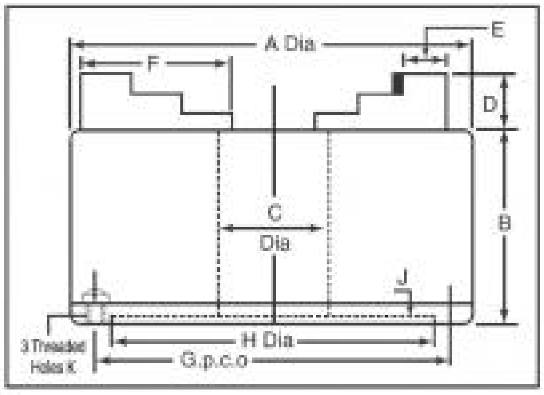
ACCURACY ADJUSTABLE (TRUGRIP) SELF CENTRING CHUCKS

TYPE AS

AS3

AS6

- Accuracy Adjustable within 0.005 mm. and repeating to within 0.013 mm.
- Toolroom Accuracy in Production.
- Costly fixtures eliminated.
- Inexpensive to install.
- Long life due to jaws theeth, scroll threads, body jaw ways, body knob hardened & precision ground



FOR APPLICATION ON

- Cylindrical & Internal Grinders
- Tool & Cutter Grinder
- Precision Lathes
- Dividing Heads

6 JAWS TRUGRIP CHUCKS

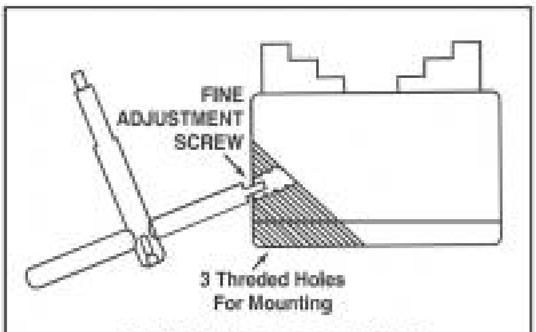
- Soft Metals Gripped firmly without undue marking.
- Fragile Components such as rings and tubes to be held Securely with Less Pressure and minimum distortion.

DIMENSIONAL DATA

Nominal Size	A	B	C	D	E	F	G	H H7	J	K	Weight. in kg	
											3 JAW	6 JAW
80	80	56	16	12	10.5	32	67	56	3	M6	2.5	-
100	100	65	20	18	14	41	83	70	3	M8	4.5	-
125	125	75	33	21	15	48	108	95	4	M8	6.0	7.0
160	160	85	42	29	18	65	140	125	4	M10	10.5	11.5
200	200	95	55	31	22	80	176	160	4	M10	13.0	17.5
250	250	110	76	36	28.50	98	224	200	5	M12	34.5	38.0
315	315	117	107	36	28.50	118	286	260	5	M14	45.0	48.0

ADJUSTING PROCEDURE

Hold the workpiece in this chuck in normal manner. Slacken the adjusting screws with dual purpose key provided. Place the dial indicator against workpiece and rotate the chuck till lowest reading is obtained. The adjusting screws nearest to low reading is then tightened until gauge needle moves half the total eccentricity reading. The same procedure is repeated till the desired level of concentricity is achieved. All adjusting screws then finally be checked for full tightness. No further adjustment is required for repetitive machining of similar jobs of same dia whichis possible within 0.013 mm accuracy. Always use NOMINATED PINION for reptitive gripping accuracy.



ADJUSTING PROCEDURE

LIMITATION:

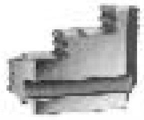
Trugrip Chucks are slightly unbalanced due to adjustment feature and may cause spindle vibration at high speed.

TECHNICAL FEATURES

BODY :

High strength cast iron alloyed with maganese and chrome (SG Iron optional)

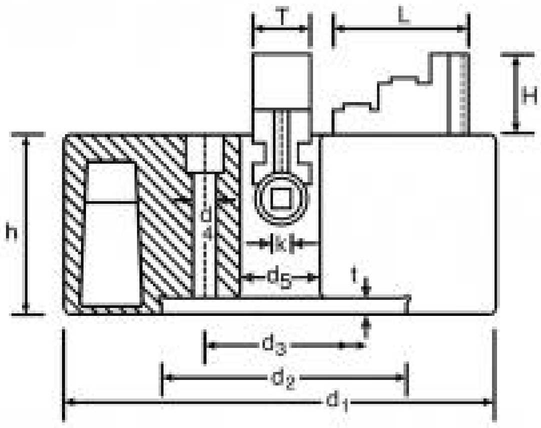
JAWS :

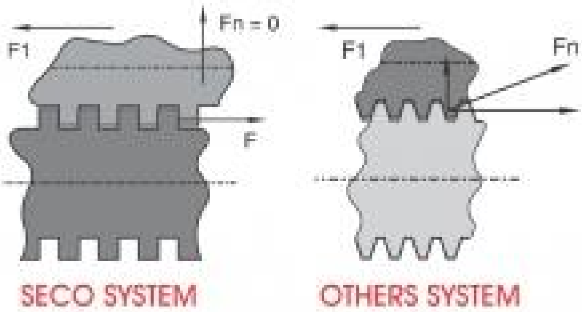


Case hardened and precision ground at gripping faces and jaw ways.

OPERATING SCREW :

- Case hardened alloy steel for strength and wear resitance.
- Squaire thread for **HIGH GRIPPING FORCE**





F : Force applied
Fn : Force lost
F1 : Gripping Force

SECO SYSTEM F1=F

Others System F1=F-Fn
F1<20%F

THURST BEARING :

Case hardened alloy steel so designed to provide maximum strength and life to working parts

DIMENSIONAL DATA (Heavy Duty)

Nominal Size d_1	d_2	d_3	d_4	d_5	t	h	k	T	L	H	Max. Rec. rev/min	Clamping Capacity		Wt. in kg
150	125	140	M-10	45	6	66	8	24	55	22	750	6	150	10
200	95	80	M-10	50	6	83	10	29	75	31	750	12	200	17
250	125	105	M-12	60	8	81	10	29	85	34	600	18	250	20
305	160	130	M-12	75	10	90	12	34	100	44	600	18	305	34
350	160	130	M-12	80	12	93	12	34	110	44	600	18	350	40
400	200	171	M-16	100	14	100	14	40	124	60	600	25	400	60
450	200	171	M-16	100	14	100	14	40	124	60	600	25	450	75
500	260	235	M-20	127 / 177	10	110	14	47	150	100	450	55	500	107
630	260	235	M-20	150/177/355	18	110	14	47	150	100	450	55	600	140
800	655/385	603/330.2	M-18/M-22	535/230/270	18	165	22	70	208	100	400	100	800	350
1000	406.4	377.82	M-25	177	18	203	16	60.32	222	100	400	100	1000	450

DIMENSIONAL DATA (Light Duty) (For Grinding Machines)

◆100	88.9	76.2	M8	25.4	2.5	35	5	12	35	16	2
▲ 150	82.5	69.8	M8	40	3	45	7	15.7	51	20	4.5
▲ 200	95.25	82.5	M10	45	3.5	50	7.5	19	63	24	8.5

◆ Steel Body ▲ SG Iron Body

FIE FOUNDATION AWARD
IMTEX 2005

SECO Dust Proof Scroll Operated Self Centering Chuck is the result of continuous Research & Development by our team committed to solve chucking problems.

SECO Dust Proof Chucks are boon for mass production workshops where dust causes periodic breakdown in chucks, thus reducing productivity. This new design of scrolloperated self centering chuck has serrated base jaws mountable with reversible Top Hard Jaws or Top Soft Jaws.(Both Top Hard Jaws and Top Soft Jaws are standard accessories with the chuck. The serrated base jaws have a limited movement and are also sealed against any dust entry into the chuck.Therefore the life of the Dust Proof SECO Chucks is increased by 60% and also the loss in productivity because of essential periodic cleaning of chuck is eliminated. This saving in terms of increased productivity and increased life of chuck makes SECO Dust Proof Chuck an indispensable equipment for any mass production workshop.

The limitation of limited movement of Base Jaws is overcome by serrations on whichthe Top Jaws can be shifted inward and outward to hold any diameter with in the permissible gripping range of Dust Proof Chucks.

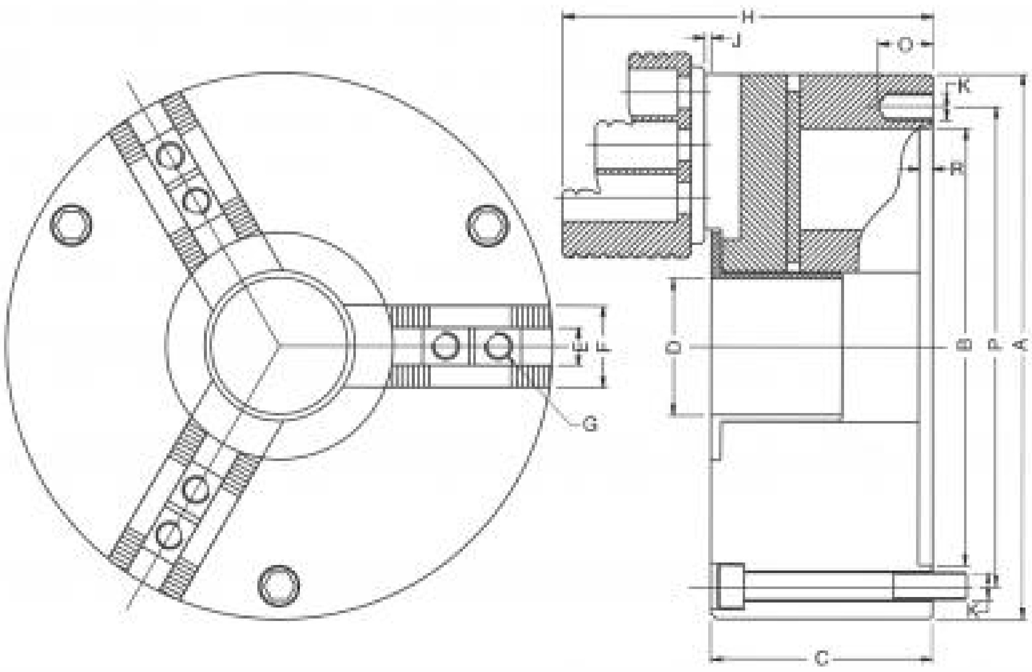
Salient Features :

BODY : SG Iron body with hardened and ground Body Jaw Ways and Body knob.

SCROLL : Alloy steel case hardened upto 45 HRc Scroll with ground thread flanks from both sides.

JAWS : Top Hard Jaws and Base Jaws are made of alloy steel, Case Hardened up to 60 HRc. Jaws teeth ground from both sides. Jaws serrations are also ground.

PINION : Pinion made of alloy steel Case Hardened up to 50 HRc.



DIMENSIONAL DATA

A	B H7	P	C	D	E H7	F	G	H	J	K	O	R	Weight
160 mm	125	140	71	37	11	27	M8 x 1.25	118	2	M10	20	4	10.0
200 mm	160	176	81	50	12	30	M10 x 1.5	136	3	M10	20	4	18.0
250 mm	200	224	90	70	17	35	M12 x 1.75	148	3	M12	24	5	30.0
315 mm	260	286	100	100	17	40	M12 x 1.75	160	3	M12	32	5	-

Serrations : 1.5mm x 60°

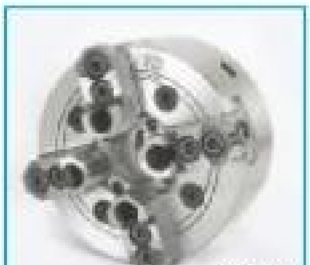
OPEN CENTRE / HOLLOW LARGE BORE DIRECT MOUNTING POWER CHUCK

TYPE OCDK

SECO
POWER

3JAW

2JAW

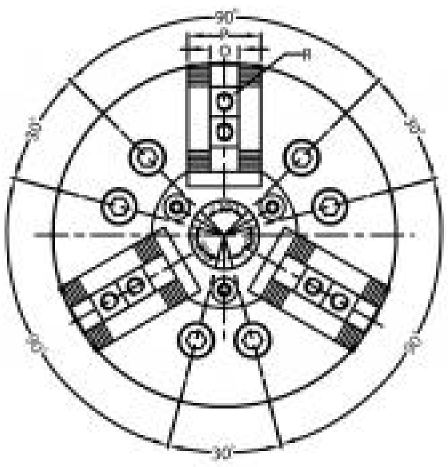


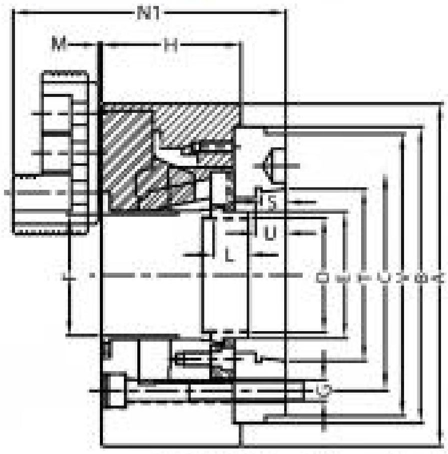
4JAW

SECO open centre wedge hook chucks with thru hole are ideal for high speed chucking, bar chucking and universal machining. These chucks come fitted with back plate ready to mount on the machine.

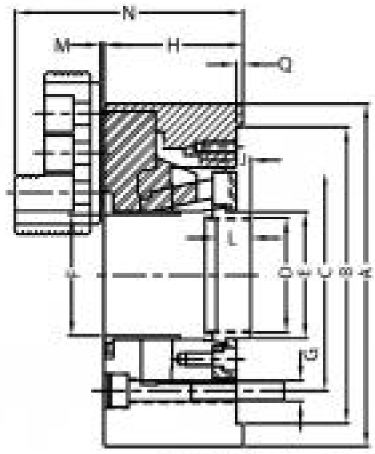
SPECIAL FEATURES :

- High Speed
- Constant accuracy & high endurance
- Strong Gripping
- Balanced
- LARGE BORE
- Available in 2, 3 & 4 Jaw configurations





OCDK (Direct Mounting)



OCK (Plain Back)

A	B	Taper Nose	C	D	E	F	G	H	J Min.	J Max.	L	M	N	N1	O H7	P	Q	R	S	T	U	Serrations
135 mm	110	A2-4	82.6	M36 x 2.0	48	33	3xM10	61	-2.30	8.20	20	1.5	92	105	10	27	4	M8 x 1.25	10	65.513	12	1.5 x 60°
170 mm	140	A2-5	104.8	M55 x 2.0	60	45	6xM10	81	-11.30	2.60	20	2	117	130	12	32	5	M10 x 1.25	12	82.563	15	1.5 x 60°
210 mm	170	A2-6	133.4	M60 x 2.0	66	52	6xM12	92	-11.70	2.25	20	2	144	163	14	37	5	M12 x 1.75	13	106.375	16	1.5mm x 60°
254 mm	220	A2-8	171.4	M85 x 2.0	94	75	6xM16	100	-10.00	9.00	20	2	156	175	16	42	5	M12 x 1.75	14	139.719	18	1.5mm x 60°
305 mm	220	A2-8	171.4	M90 x 2.0	108	90	6xM16	110	-10.00	9.00	20	4	161	185	18	50	6	M14 x 2.0	14	139.719	18	1.5mm x 60°
380 mm	300	A2-11	235.0	M130 x 2.0	139	117.5	6xM20	131	-10.50	14.50	34	5	222.5	250	24	62	6	M20 x 2.5	16	196.869	20	1.5mm x 60°
450 mm	380	A2-11	235.0	M130 x 2.0	139	117.5	6xM20	131	-10.50	14.50	34	5	222.5	250	24	62	6	M20 x 2.5	16	196.869	20	1.5mm x 60°
530 mm	380	A2-15	330.20	M155 x 3.0	170	140	6xM22	140	-11.50	13.5	41	6	231.5	276.5	24	62	6	M20 x 2.5	17	285.78	21	1.5mm x 60s

EXTRA LARGE BORE

A	B	Taper Nose	C	D	E	F	G	H	J Min.	J Max.	L	M	N	N1	O H7	P	Q	R	S	T	U	Serrations
170 mm	140	A2-5	104.8	M60 x 2.0	65	52	6xM10	81	-11.50	1.00	20	2	117	130.0	12	32	5	M10 x 1.25	12	82.563	15	1.5 x 60°
210 mm	170	A2-6	133.4	M75 x 2.0	80	62	6xM12	91	-17.20	3.00	20	2	144	163.0	14	37	5	M12 x 1.75	13	106.375	16	1.5mm x 60°
254 mm	220	A2-8	171.4	M98 x 2.0	123	90	6xM16	100	-10.50	11.00	20	2	156	175.0	16	42	5	M12 x 1.75	14	139.719	18	1.5mm x 60°
325 mm	300	A2-11	235	M115 x 2.0	123	117	6xM20	112	-5.50	14.50	34	3	169	193.0	21	52	6	M14 x 2.0	16	196.869	20	1.5mm x 60°

NOTE : 1/16" x 90° serrations are optional.

SIZE	Max. RPM	Jaw Stroke (Per Jaw)	Plunger Stroke (Max.)	Weight (Kgs.) approx	Operating Force (Max. Kgf.)	Gripping Force (Max. Kgf.)
135 mm	6000	2.75	10	9	1600	3000
170 mm	6000	2.75	12	12	3000	7000
210 mm	5000	3.75	16	20	3000	7500
254 mm	4200	4.4	19	35	5000	10000
305 mm	3200	4.4	19	50	6000	13500
325 mm	3200	4.4	20	65	6000	13500
380 mm	2500	5.3	25	120	7200	18000
450 mm	2200	5.3	25	130	7200	18000
530 mm	2200	5.3	25	130	7200	18000

Chucks can be supplied with back plates of other International Standards on demand

KITAGAWA B200 Compatible